AUS | الجامعة الأمريكية في الشارقة
American University of Sharjah

Department of Mathematics and Statistics

MTH 343

Homework 2 - Fall 2024

The homework must be submitted to ybelhamadia@aus.edu by Thursday November 14th 2024 at 23 :59.

1. Using some provided data set, we need to build the Lagrange interpolating polynomial $L(x)$. We want to evaluate $L(x)$ at a given point x_p, i.e. we want to find $y_p = L(x_p)$. Write a MATLAB function that implements Lagrange Interpolation Polynomial algorithm. The interpolating polynomial is passing through the data points :

$$P_1 = (x(1), y(1)), P_2 = (x(2), y(2)), \cdots, P_N = (x(N), y(N)).$$

The input parameters of the MATLAB function must be the vector x, the vector y and x_p (x_p is the value where we want to evaluate $L(x)$). The output must be the computed value y_p for interpolation ($y_p = L(x_p)$).

2. Write a MATLAB function that implements the Natural Boundary Cubic Splines algorithm (given below). The input parameters must be the vector x of values for the nodes, and the vector $y = f(x)$. The output of the MATLAB function must be a matrix containing the coefficients of the spline polynomials for each interval.

The Natural Boundary Cubic Splines $S(x)$ is given by the following algorithm :

(a) Input : $x_0 \cdots x_n$; $a_0 = f(x_0), \cdots, a_n = f(x_n)$

(b) Output : a_j, b_j, c_j, d_j for each $j = 0 \cdots n-1$

(c) For $j = 0 \cdots n-1$ set $h_j = x_{j+1} - x_j$

(d) Set the matrix $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & & & \\ h_0 & 2(h_0+h_1) & h_1 & & \\ & \ddots & \ddots & \ddots & \\ & & h_{n-2} & 2(h_{n-2}+h_{n-1}) & h_{n-1} \\ & & & 0 & 1 \end{bmatrix}$$

(e) Set the vector $\mathbf{z}$

$$\mathbf{z} = \left(0, \frac{3(a_2-a_1)}{h_1} - \frac{3(a_1-a_0)}{h_0}, \ldots, \frac{3(a_n-a_{n-1})}{h_{n-1}} - \frac{3(a_{n-1}-a_{n-2})}{h_{n-2}}, 0\right)^T$$

(f) Solve **Ax=z** to obtain $\mathbf{x} = (c_0, \ldots, c_n)^T$

(g) Set $b_j = \dfrac{1}{h_j}(a_{j+1} - a_j) - \dfrac{h_j}{3}(2c_j + c_{j+1})$, and $d_j = \dfrac{1}{3h_j}(c_{j+1} - c_j)$.

(h) Output : a_j, b_j, c_j, d_j for each $j = 0 \cdots n-1$

$$S(x) = \begin{cases} S_0(x) = a_0 + b_0(x - x_0) + c_0(x - x_0)^2 + d_0(x - x_0)^3 & x \in [x_0, x_1] \\ S_1(x) = a_1 + b_1(x - x_1) + c_1(x - x_1)^2 + d_1(x - x_1)^3 & x \in [x_1, x_2] \\ \vdots & \vdots \\ S_{n-1}(x) = a_{n-1} + b_{n-1}(x - x_n) + c_{n-1}(x - x_{n-1})^2 + d_{n-1}(x - x_{n-1})^3 & x \in [x_{n-1}, x_n] \end{cases}$$

3. Create a script to test the above two codes. You need to plot the points and the curves obtained by both the spline and Lagrange interpolating polynomial in the same figure. Use the following data set :

x	0.9	1.3	1.9	2.1	2.6	3.0	3.9	4.4	4.7	5.0	6.0	7.0	8.0	9.2	10.5	11.3	11.6	12.0	12.6	13.0	13.3
$f(x)$	1.3	1.5	1.85	2.1	2.6	2.7	2.4	2.15	2.05	2.1	2.25	2.3	2.25	1.95	1.4	0.9	0.7	0.6	0.5	0.4	0.25

4. Your final submission should include :

   (a) **(4 pts)** Your M-file function for question 1)
   (b) **(4 pts)** Your M-file function for question 2)
   (c) **(2 pts)** Your M-file script for question 3)